Why adding uncertainty into regional GVA is a good thing

Dan Olner

2026-02-11

Table of contents

Things to stick in

- Make clear this is focused on thinking through regional econ strategy and what difference it makes for that specifically.

Headlines

-
- This is in large part due to the constrictions of the (globally agreed so extremely slow to change) System of National Accounts (Coyle 2014).
- Far from just introducing less precision, including uncertainty leads to some quite different, useful ways of thinking about regions, grounded in what we do and don't know.

- I might argue - it's more powerful, because it short-circuits many unhelpful ways of thinking about growth and industrial strategy (some built right into the foundation e.g. see the CIA Russia assessment).

Resources/links

I have prepared two interactive web pages for each of the GVA sources discussed below. I'll refer back to these with links.

1. Chained volume GVA: What is the growth signal if error rates are included in regional data?
2. Current prices GVA: What happens to location quotients if error rates are included?

Introduction: how to avoid chasing vampires off cliff edges

In the movie 'The Lost Boys', David and his west-coast vampire buddies race motorbikes across a misty landscape. Michael (as yet unaware of the blood-drinking habits of his new associates) tries to keep up with them. They egg him on - but then he spots the faint beam of a lighthouse, puts two and two together and barely avoids hurtling over a cliff edge.

One might say David was claiming 'incredible certitude' (Manski 2020) about the cliff's location. "Don't worry Micheal, it's definitely, definitely miles from here."

If there's fog, what you do about that depends on what you think is going on / what else you feel certain about. But the fog matters, the fog itself is information. (i.e. fog near a cliff is different info to fog walking on an open moor).

Being constructive in a noisy argument

Put this in resources as its own thing/post. Would allow discussion of 'concern trolling' a bit?

Getting away from perfect sums

The issues all stem from the same root: national accounts methods (1) are designed to analyse nations, not regions, and (2) are accounts. They have to balance, every part of the economy has to add up correctly and consistently - including across all UK regions. This forces a national-level, top-down straitjacket on regional numbers. Let's make a start in this post on one of the most important consequences of that:

- Accounts only report single values. It's all they *can* report. Uncertainty doesn't feature.

Through this lens, the UK grew by exactly 0.1% in the three months to November 2025. Uncertainty only enters via later revisions to those exact figures - the image here is from the ONS showing how much revisions over time differ from initial numbers. These are very much *not* confidence intervals showing the underlying uncertainty in the data.

And they try to do this work when chancellors and shadow chancellors shout and point fingers over every 0.1% revision. Argh.

Or: the ONS have no choice here - the national accounts straitjacket…

In the ONS' method doc for regional GVA, they explicitly say:

> "The complex process by which GVA estimates are produced means that it is not currently possible to define the accuracy of the estimates… for example, through their standard errors. Therefore, the reliability of the estimates is measured by the extent of revisions."

I think ONS are doing themselves a disservice here - they could certainly handle the complexity. It's just the 'accounting' point again: the assembly of national output involves so many moving parts, and the need for perfect balancing (however baroque) so built-in, there just isn't scope to do anything else with regional economic numbers.

The uncertainty, however, hasn't gone away. In this post, I'll have a go at showing what difference the uncertainty could make to the numbers, if we could see it.

That's a different (though of course linked) question from "If things are uncertain, how does that change how we should act?" I'll come to that one in a later post (I'm just reading Manski's Lure of Incredible Certitude to warm up for that).

This is all a slightly alarming look back at the certitude I've built into my own work too. It's very difficult to resist, both methods-wise (single values is all we have, often) and what-policymakers-want-to-see-wise. One of my plots below (code here) is a prime example: GVA per hour, ITL2 zones ranked! Who's up, who's down on the scoreboard??

Making a data-driven guess about the level of regional GVA uncertainty

The **Annual Business Survey (ABS)** is a good source that allows an educated guess of confidence bounds around regional/sectoral GVA data. It is designed to capture sectoral and regional GVA at a reasonable resolution. As the ONS say in their fascinating breakdown (if you're into that kind of thing) of "observed/estimated/modelled components of regional GVA":

> Of all the data sources used in regional GVA, the ABS has the greatest overall impact, representing around 71% of GVA(P) and 22% of GVA(I)… It includes elements corresponding to all three of the categories of data we wish to analyse: directly

collected from businesses operating in a single region; weighted to represent non-sampled businesses; and apportioned to regions from UK-wide company information.

Data comes direct from firms - there is much less reliance on some of the more heroic assumptions that go into other aspects of regional GVA. [Roadmap: exploring the implications of assumptions for finance, telecoms and public sectors].

The publically available ABS sources are broken into the central estimate data and error data[1] giving GVA values and standard errors at ITL1 geography level and 2-digit SIC sectors. Here, I convert GVA and standard errors from the ABS to coefficients of variation (proportion of error) and then apply them to the regional/sectoral GVA national accounts data at the same geogaphical scale, and where sectors are present in both sources (there are 73 that match once ABS sector categories are processed to match the region/sector data's bespoke SIC list). Only years available in both sources are used - 2012 to 2023.

Below, I'll explore the implications of these error rates when applied both available types of GVA data:

- Chained volume (CV) data that aims to capture 'real' output changes over time. From these, we can ask, "Did this sector grow or shrink here?" We can do a few other things e.g. was its growth slope really steeper than other places?
- Current prices (CP) data, which (being prices as they were in the year they were counted) can be summed, and used to look at location quotients (LQs) - how concentrated a sector is in each place. It's where we can get a good sense of the UK's economic structure, because it's possible to compare sizes across the UK. We'll have a go at adding error rates to LQs.

For both of those, the question is: "If the regional GVA data had the same error rates as the ABS, what would it look like and what would some implications for analysis be?"

Are those reasonable questions to ask? This is asking, "If the error rates were this, then…" For those sectors that match, it's a fair assumption that the ABS reasonably captures its uncertainty. The error may be smaller or larger, but including it is almost certainly more accurate than the alternative - working with the point data as if it was the single true value.

Seeing what difference error makes - even if there's error in our error - leads to quite different ways of thinking.

Chained volume GVA: growth over time

Chained volume data is used to assess real economic change over time, accounting for inflation and quality changes in goods and services. Separate deflators are calculated for each sector

[1] code/comments here run through combining those two sources. A final joined CSV is here, to save others the pain, though it's just for GVA, not every ABS value.

within each region, with a single year set as the point where the chained volume and 'prices that year' amounts are the same. Other years are adjusted relative to that, for each sector/place combination. (As a result, each sector/place time series must be treated separately and cannot be summed.)

Change in CV values over time should represent actual growth or shrinkage in a particular sector and place. This section limits itself to asking how much the growth signal changes for individual sectors in specific ITL1s with the inclusion of error rates.

[It doesn't use those errors to compare sectors across places… though might do this quickly… simulation again to add to CAGR]

This interactive web page provides a way to look at individual sectors across all twelve ITL1 zones.

- The inclusion of error rates immediately makes it very rare for year-to-year changes to be distinguishable from zero. So instead, what each grid plot does is compare each year to every other year, to show where each is clearly separable.
- The assumptions used (on top of the if/then above) are: 95% confidence intervals, and assuming that GVA change isn't clearly present if the possible minimum from one year is lower than the possible maximum from another year (and vice versa). This is a conservative assumption that the true value[2] could be at both confidence extremes between timepoints.

Let's talk through the default sector in the interactive, fabricated metals, to make more sense of that. Consider the example in Figure 1 - in the interactive, hover over Yorkshire and Humber to see its growth over time. What this shows:

- Blue squares in the grid indicate that the column year was clearly higher than the row year it is compared to ('clearly' as described above).
- We can see that 2020 to 2023 are clearly higher than the earlier years in the data, from 2012 to 2016, though the high peak in 2021 is the only year consistently separable from others. 2020 to 2023 are years where the confidence interval minimum is separable from the earlier CI maxima.
- The rest of the grey area is showing that, for the first half of the data from 2012 to 2017 - on the assumptions we have here - no clear growth signal is coming through.
- Note, the grid is mirrored along the diagonal: here, orange squares are showing the same thing in reverse - earlier column years are clearly lower than some later ones.

Scotland shows the opposite pattern, which can be seen from its growth slope in the interactive clearly too - bottom left orange squares indicate clearly lower values in the later part of the

[2] 'True value' being used as shorthand for the cumbersome '95% confidence interval would contain the true value 95% of the time if the survey sample was taken again'. [ONS source…]

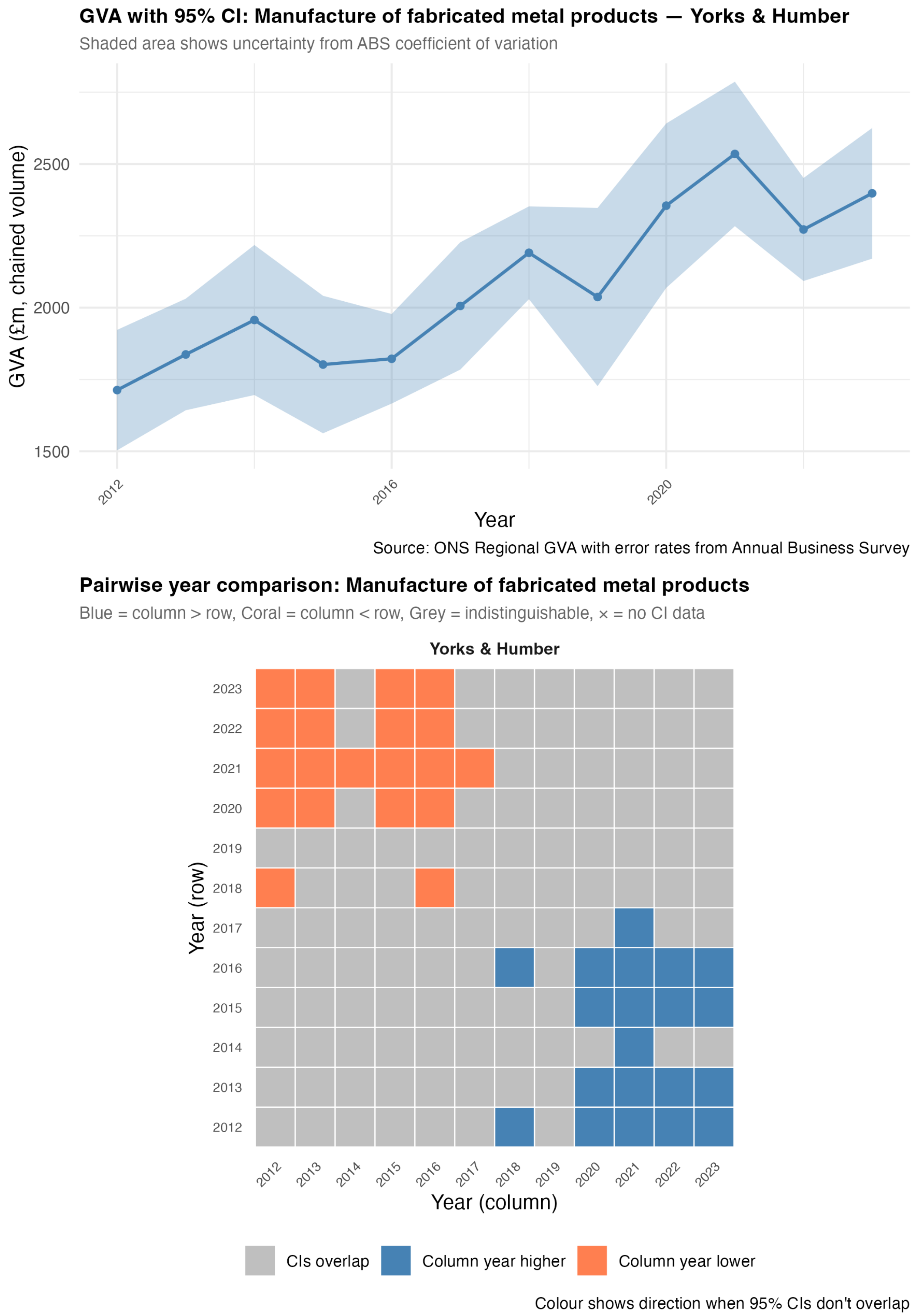

Figure 1: Yorkshire & Humber GVA growth grid and growth over time, fabricated metals.

data. Figure 2 is showing that, while the later value increases appear separable from the low point around 2017, it can't be so easily distinguished from where it was from 2012 to 2016, despite the central estimates looking higher (they could well have been higher in the earlier period and lower in the later one). Several other ITLs are purely grey. Looking at any of those shows combinations of wide error rates and fairly flat change over time, considering the range of those confidence intervals. The apparent drop in the North West after 2017, for example, isn't clearly separable, though it looks strong. Other sources could help support or question that apparent change.

Within this way of looking, the impact of the COVID19 pandemic on land transport is very visible. Mid-pandemic drops show up as an orange 'column year lower than most other years' band. London stands out as the one place that hasn't yet seen a bounce-back, remaining significantly lower than other ITL1s.

Other notable patterns are discussed in a section at the end of this document.

Discussion:

The binary cutoffs here are stark - this is a common issue with threshold-based statistical approaches… discuss pros and cons and say still probably better than the alternative. Consider these as part of decision-making under uncertainty. What does that look like?

Arguments about where to put the level

But not limited to single one, and there isn't a single 'correct' answer. The approach above can pull out very clearly strong growth signals. We could explore less conservative assumptions and compare to other sources. We could also use other methods - for example, there is likely information in the correlation between timepoints in Figure 2's latter high values that could change the picture.

None of that changes the fundamental point: introducing uncertainty doesn't have to mean losing information. In fact it can be the reverse - revealing genuine signal in the noise.

That even with large error ranges, the signal you get is highly valuable

And arguably the risk of false positives is less than the opposite.

Current prices GVA: location quotients

Location quotients (LQs) are an intuitive as easy to calculate way to assess the relative strength of sectors in regions. It is the ratio of a sector's proportion of a region versus that sector's

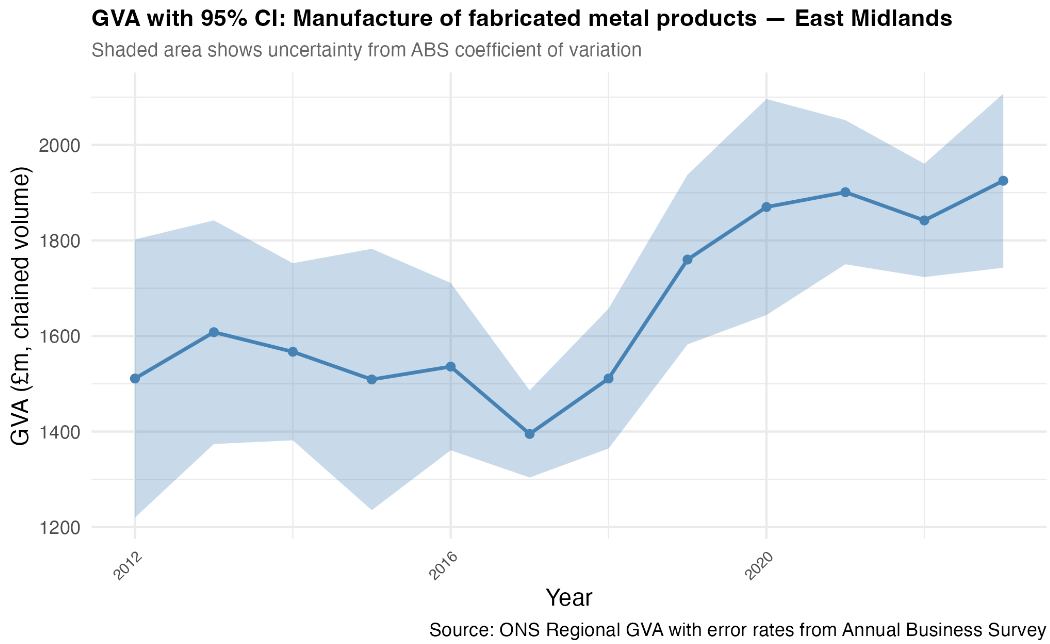

Figure 2: East Midlands GVA growth grid and growth over time, fabricated metals

proportion of the UK as a whole. If, say, fabricated metals is 20% of a sub-region's GVA but only 10% nationally, it is twice as concentrated - the LQ is two[3].

Current price (CP) GVA data can be used for LQs because (unlike CV) it *can* be summed, as it is just the money value of each sector's output recorded in that year. CP data can't show real growth over time, but it can be used to examine economic structural change - how much a sector/place combination has *relatively* grown in concentration (which could be due to growth in that sector or other sectors increasing in nominal value faster).

Location quotient

$$\mathrm{LQ}_{s,r} = \frac{\mathrm{GVA}_{s,r} \;/\; \mathrm{GVA}_{r}}{\mathrm{GVA}_{s,\mathrm{UK}} \;/\; \mathrm{GVA}_{\mathrm{UK}}}$$

where $\mathrm{GVA}_{s,r}$ is the GVA of sector s in region r, GVA_{r} is total GVA across all sectors in that region, and $\mathrm{GVA}_{s,\mathrm{UK}}$ and $\mathrm{GVA}_{\mathrm{UK}}$ are the corresponding UK-wide values. An LQ above 1 means the sector is more concentrated in that region than nationally; below 1, less so.

LQs are (in theory) perfect for thinking about regional specialisms, and to comparison and ranking, making them an obvious fit for industrial strategy thinking. But they also carry the "single accurate value" issue over from the underlying GVA data. One single LQ value exists per sector/place combination in any year.

Including error rates around an LQ is trickier than for direct GVA because it is a derived value - it doesn't make sense to apply confidence intervals to an LQ directly. The solution used here is to simulate what the range of the underlying GVA values could be, and then calculate LQs based on those simulated values (each a normally distributed range based on the standard errors applied from the ABS, adjusted to make sure no negative values arise as these could break the LQ formula). This is then repeated 500 times to produce a range of possible LQs, built on a spread of likely underlying GVA numbers.

Here's the interactive page for looking at the results, starting again with fabricated metals LQs for each ITL1 and including 95% confidence intervals (see also Figure 3). Considering overlapping error rates, East and West Midlands and Yorkshire & the Humber might be seen as one 'high concentration' group not really separable from each other, with London and the South East separately at the 'low concentration' end. West Midlands also appears to have clearly the highest concentration. Other ITL1s are not so separable, and four have error bars crossing zero - if these are reasonable, there is no way to say if fabricated metals is more or less concentrated than average.

[3]because (A/B)/(C/D) is equivalent to (A/C)/(B/D), the LQ actually captures two related ways of seeing the same thing: how relatively concentrated sectors are *across a whole geography* like the UK, and how concentrated *within a subgeography* like South Yorkshire they are. (See the desciption and table in the ONS' older LQ document.)

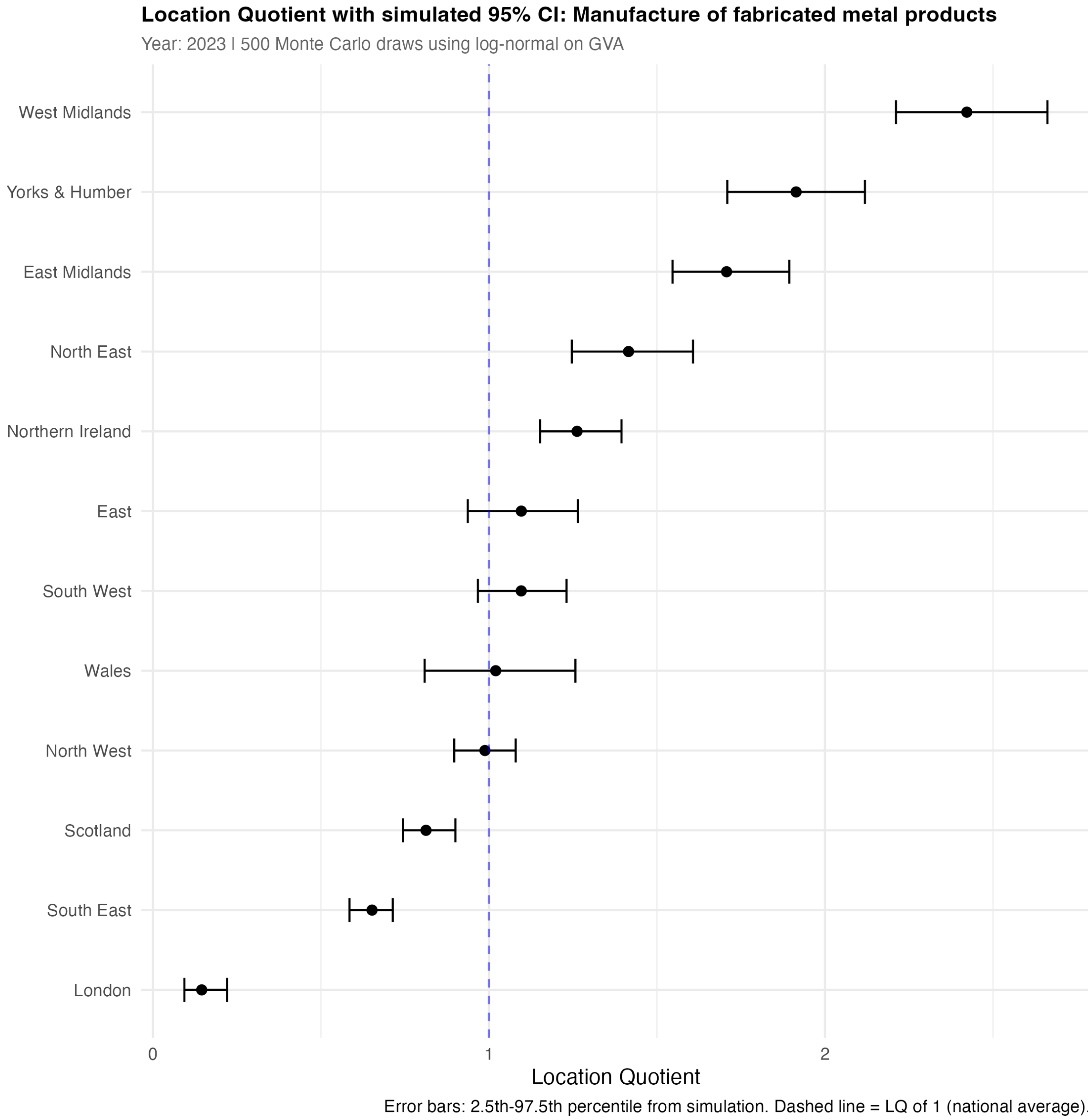
Location Quotient with simulated 95% CI: Manufacture of fabricated metal products
Year: 2023 | 500 Monte Carlo draws using log-normal on GVA
West Midlands
Yorks & Humber
East Midlands
North East
Northern Ireland
East
South West
Wales
North West
Scotland
South East
London
0
1
2
Location Quotient
Error bars: 2.5th-97.5th percentile from simulation. Dashed line = LQ of 1 (national average).

Figure 3

The plot also has an option to compare LQs across five years, between 2017 and 2023. If overlapping error bars suggest the LQ for fabricated metals may not have significantly changed in many places, the East Midlands (real LQ increase), Scotland and the South East (a plausible decrease in both) do suggest real change.

Claude-code edits and additions

- Claude code added the LQ formula and the lines from "where GVA…" to "below 1, less so" in Current prices GVA: location quotients.

Coyle, Diane. 2014. *GDP: A Brief but Affectionate History.* Princeton University Press.

Manski, Charles F. 2020. "The lure of incredible certitude." *Economics and philosophy* 36 (2): 216245. https://doi.org/10.1017/S0266267119000105.